

1. Overview

This document captures key learnings, challenges, and decisions made during the analysis and design of the **E-commerce Sports Shop System**.

The purpose of this document is to reflect the BA's understanding, reasoning, and professional growth throughout the project lifecycle.

2. Key Learnings

2.1 Understanding Business vs System Responsibility

During this project, I learned how to clearly distinguish between **user actions** and **system responsibilities**.

- Customers initiate actions such as selecting products, adding items to cart, and confirming orders.
- The system is responsible for validating data, checking business rules, processing orders, and generating confirmations.

This distinction helped improve the clarity of **Activity Diagrams** and **Sequence Diagrams**, ensuring correct swimlane usage.

2.2 Importance of Scope Definition

Defining **In Scope** and **Out of Scope** early helped prevent over-design.

For example:

- Online payment gateway integration was intentionally marked as **Out of Scope**.
- As a result, payment processing was excluded from activity and sequence diagrams.

This reinforced the BA principle:

"Not everything that can be built should be built in the first version."

2.3 Business Rules vs Functional Requirements

I learned to separate **Business Rules** from **Functional Requirements**:

- Functional Requirements describe *what the system does*.
- Business Rules describe *constraints and policies* that govern behavior.

This separation improved traceability and made the requirements easier to maintain and review.

3. Challenges Faced

3.1 Designing Correct UML Relationships

One of the main challenges was choosing the correct relationship types in the **Class Diagram**:

- Association
- Composition
- Cardinality (1, 0..1, 0..*)

Understanding real-world ownership helped:

- Cart **composes** CartItem
- Order **composes** OrderItem
- Product is referenced but not owned

3.2 Handling Alternative Flows in Sequence Diagram

Modeling alternative flows such as:

- Empty cart during checkout
- Invalid login credentials

required careful use of **alt** blocks to ensure:

- The system does not proceed without user interaction
- Error handling flows are explicit and realistic

4. Design Decisions Made

4.1 Login Trigger Point

Login is required **only when the customer checks out**, not when browsing products.

Reason:

- Improves user experience
- Aligns with common e-commerce practices

4.2 Single Active Cart per Customer

Each customer is allowed to have **only one active cart at a time**.

Reason:

- Simplifies cart management
- Reduces system complexity

4.3 Order Creation Timing

An order is created **only after checkout confirmation**, not when items are added to the cart.

Reason:

- Prevents unnecessary order records
- Aligns with real business workflows

5. What I Would Improve in the Next Iteration

If this project were extended, the following enhancements could be considered:

- Payment gateway integration
- Order cancellation workflow

- Delivery tracking
- Promotional pricing rules

These features were intentionally excluded to keep the project focused and aligned with scope.

6. BA Skills Developed

Through this project, I improved the following BA skills:

- Requirement analysis
- UML diagramming
- Business rule identification
- Scope management
- Communication through documentation

7. Interview Talking Points

This project enables me to confidently answer:

- How I analyze business problems
- How I transform requirements into diagrams
- How I collaborate with stakeholders
- How I manage scope and assumptions

8. Conclusion

This project helped solidify my understanding of the **end-to-end BA process**, from business context to detailed system design.

It reflects both analytical thinking and structured documentation aligned with industry standards.